ARYAN ROUTRAY

Electronics and Communication Engineer | VLSI Design | RTL Design | Embedded Systems | IoT

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PROFESSIONAL SUMMARY

Electronics and Communication Engineering undergraduate (B.Tech 2027, CGPA 8.76/10) seeking an entry-level ECE role in VLSI design, embedded systems, FPGA development or communication systems. Skilled in Verilog RTL design, functional verification and the ASIC design flow using Cadence Genus, Innovus, Tempus and Virtuoso, including IC layout with DRC/LVS. Develops embedded firmware in Embedded C and assembly on 8051, ARM Cortex-M4 and RISC-V, with sensor interfacing and IoT over MQTT. Author of an IEEE publication on deep Q-learning for cognitive radio spectrum management.

CORE COMPETENCIES

VLSI Design | RTL Design | Digital Electronics | ASIC Design Flow | Logic Synthesis | Physical Design | Static Timing Analysis | Functional Verification | FPGA Prototyping | Embedded Systems | Microcontrollers | Embedded C | IoT | Sensor Interfacing | Wireless Communication | PCB Assembly

EDUCATION

Vellore Institute of Technology, Chennai Aug 2023 – May 2027

B.Tech, Electronics and Communication Engineering | CGPA: 8.76/10

Maa Bharti Sr. Secondary School, Kota May 2023

CBSE Class XII

TECHNICAL SKILLS

- **Digital Design & VLSI:** Verilog HDL, VHDL, RTL design, combinational and sequential logic, finite state machines (FSM), simulation, functional verification; ASIC design flow, logic synthesis, physical design, static timing analysis (STA), IC layout, DRC/LVS
- **EDA & Simulation Tools:** Cadence Virtuoso, Assura, NCLaunch, Genus, Innovus, Tempus, IMC, Microwind, Verilator, GTKWave, Xilinx toolchain, Intel Quartus
- **Embedded Systems:** Embedded C, 8051 Assembly; 8051, TM4C123GXL (ARM Cortex-M4), Arduino, ESP32, Raspberry Pi, VEGA Aries v3 (RISC-V); GPIO, PWM, timers, I2C, 1-Wire, sensor interfacing, data acquisition
- **FPGA & PLDs:** Xilinx Spartan, Virtex, Altera Cyclone; RTL-to-bitstream
- **Communication & Networking:** Wireless communication, cognitive radio, dynamic spectrum access, Wi-Fi, MQTT, REST and WebSocket protocols
- **Electronics & Manufacturing:** PCB assembly, SMT processes, functional testing, quality inspection, transformer winding
- **Programming & Software:** Python, Java, Embedded C; FastAPI, React, SQLite, scikit-learn, deep reinforcement learning (DQN)

EXPERIENCE

Digital Design Intern | Marqueesemi India Pvt. Ltd., Bhubaneswar, Odisha Jul 2026

- Wrote and debugged module-level RTL in Verilog for digital design blocks.
- Took FPGA and ASIC design modules through simulation and functional verification.

Manufacturing Intern | Salcomp Manufacturing India Pvt. Ltd. Jun 2026

- Studied end-to-end smartphone charger manufacturing, with hands-on transformer winding and PCB assembly.
- Observed SMT lines, functional testing and quality inspection across the production floor.

VLSI & FPGA Trainee | Central Tool Room & Training Centre, MSME May 2025 – Jun 2025

- Designed IC layouts in Microwind and trained hands-on in VHDL and Verilog.
- Configured PLDs (Spartan, Virtex, Cyclone) using Xilinx and Altera Quartus toolchains.

PROJECTS

Smart ICU Patient Monitoring System | Biomedical Electronics, IoT, Embedded

- Built a real-time vitals acquisition system interfacing AD8232 (ECG), MAX30102 (SpO2/HR) and DS18B20 (temperature) sensors with an Arduino Uno R4 WiFi.
- Streamed sensor data over MQTT to a Raspberry Pi 5 gateway; FastAPI backend (REST + WebSocket, SQLite, role-based access) drives a React dashboard with live ECG waveforms and threshold alerts.

Automated Material Sorting System | TM4C123GXL (ARM Cortex-M4), Embedded C Apr 2026

- Developed a real-time embedded sorting system in Embedded C, classifying materials with a TCS3200 colour sensor.
- Implemented lighting-robust threshold detection and dual SG90 servo actuation through precise GPIO and PWM control.

RFID-Based Smart Parking System | 8051 Microcontroller, Assembly Apr 2025

- Implemented RFID vehicle authentication in 8051 assembly, with register-level servo gate control and precise timer-based timing.
- Integrated IR sensors for slot occupancy detection and displayed real-time availability on an LED panel.

PUBLICATIONS

Simulation-Based Dynamic Spectrum Management Using Deep Q-Learning for Cognitive Radios IEEE

- Designed a Deep Q-Network agent for adaptive spectrum access in cognitive radio networks.
- Added proactive fallback channel switching, evaluated under simulated primary-user activity with QoS-aware rewards.

ADDITIONAL INFORMATION

- **Strengths:** Problem-solving, analytical thinking, technical communication, teamwork, adaptability
- **Languages:** English, Hindi, Odia, Spanish